Discussion Unit:

Provisioning, (Cloud Providers)

Introduce the tools/vendors available for provisioning larger DSE environments

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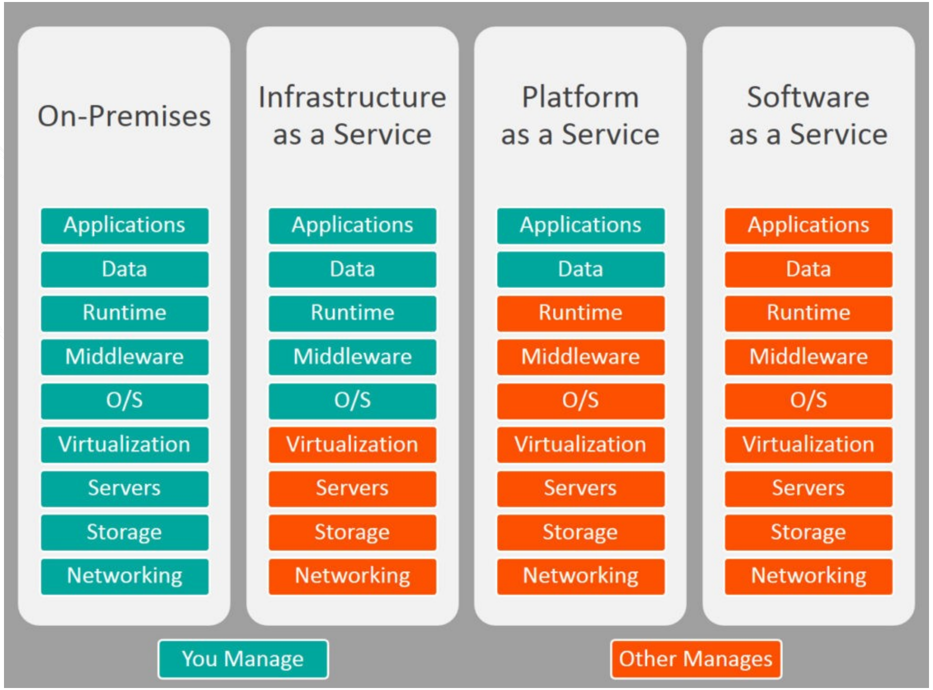
Discussion Lab:

What is ? (Define the term)



Source: http://blogs.bmc.com/wp-content/uploads/2017/09/iaas-paas-saas-comparison-1024x759.jpg

Cloud, Why: Discuss-

Source: https://idobi.com/idobi/new-xtremebitz-podcast-24/

- What changes in technology enabled the rise of cloud computing ?
- What are the benefits of cloud computing ?
- What are (n) essential characteristics of a cloud service ?

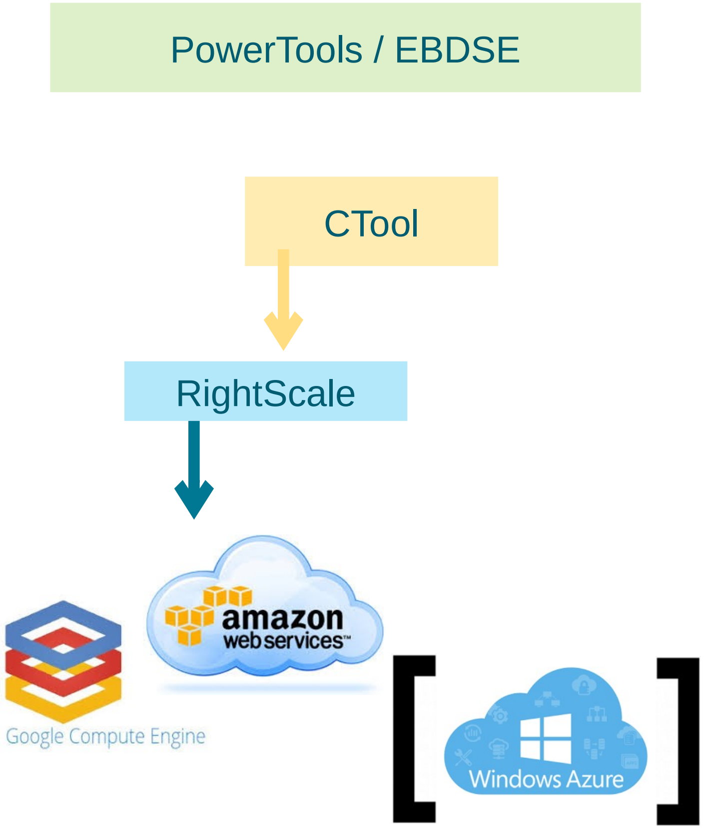
- Define and state the benefits of:
  -- IaaS
  -- PaaS
  -- SaaS

End of Discussion Lab:

Available to You:

- GCE, AWS, Azure
- RightScale
  - -- 3rd party tool
  - -- Multi-cloud provider
  - -- Give you (empty) VMs
  - -- Azure coming
- CTool
  - -- Internal tool
  - -- Uses RightScale
  - -- Gives you DSE clusters
- PowerTools/EBDSE
  - -- "Engine Block for DSE"

-- Apps, data



Just RightScale:

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RightScale Overview:

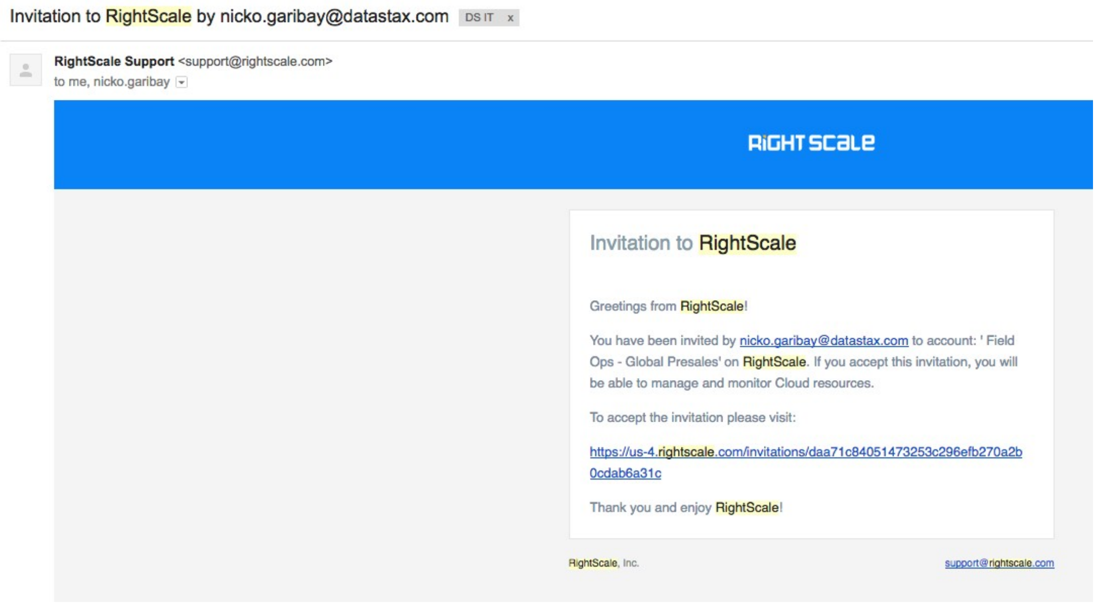
- Management layer for per person usage tracking
- Two layers of access, RightScale console and RightScale “Self Service”
- Highlevel instance / Cluster / Deployment dashboards
- Deploys on GCE or AWS
- Various teams have built some templates to utilize
  -- The workshop template is an example of one

Logging in to RightScale:

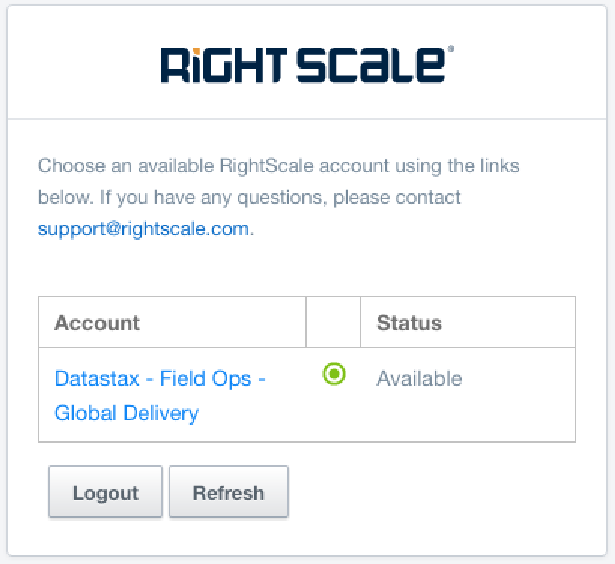
Current as of 2018/06/30-

- You should have automatically been given a RightScale account and been notified of same via email at time of hire. See image at right-
- If not, perhaps try,
  nicko.garibay@datastax.com
- If not, consult your manager

- RightScale login Url, https://login.rightscale.com/login/session/new



‘Bill-to’ modal dialog:



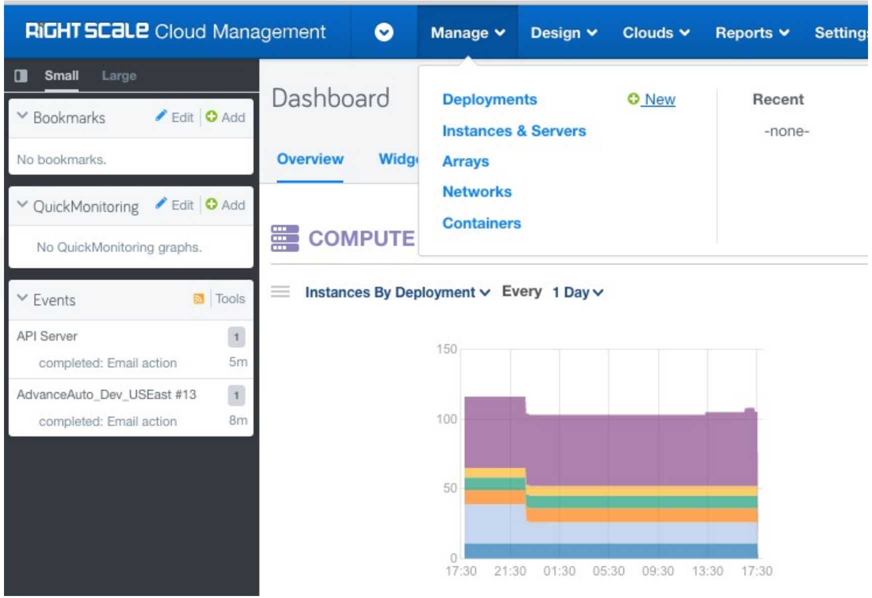
- After username password pair verification, you will receive a modal display asking you to select one of possibly several cost centers to bill your activity to.
- The display on the left displays only one such option.
- Click the correct account listing to proceed.

So again ..

- There are other installations you can create, via other methods. Using brute force (and brief time and energy), we are proceeding to create a 5 node CentOS version 7 Linux cluster in one hosted AWS availability zone.

- 6 steps, and they are-

Step 1: Deployments +New

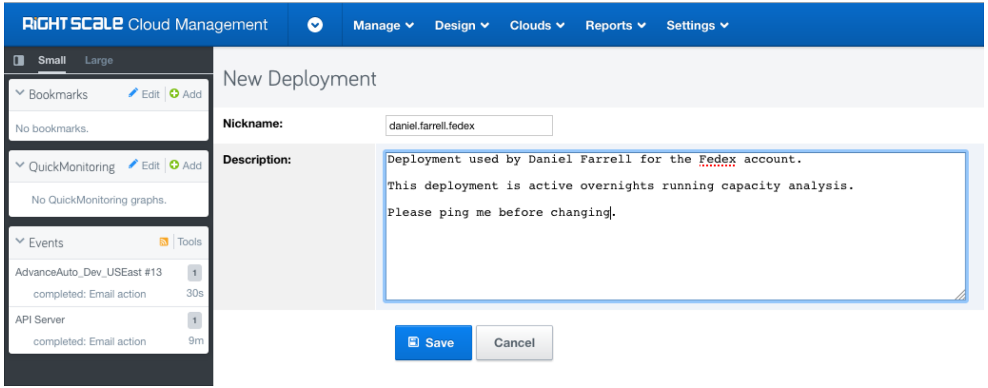


- From the Menu Bar select,
  Manage ->
  (Deployments) +New

Step 2: Identify this deployment

- Enter a Nickname that includes your DataStax identifying information, optionally a Description, and Click, Save.

- Why enter your name ? So that you are more likely to be contacted before someone else blows this deployment away.



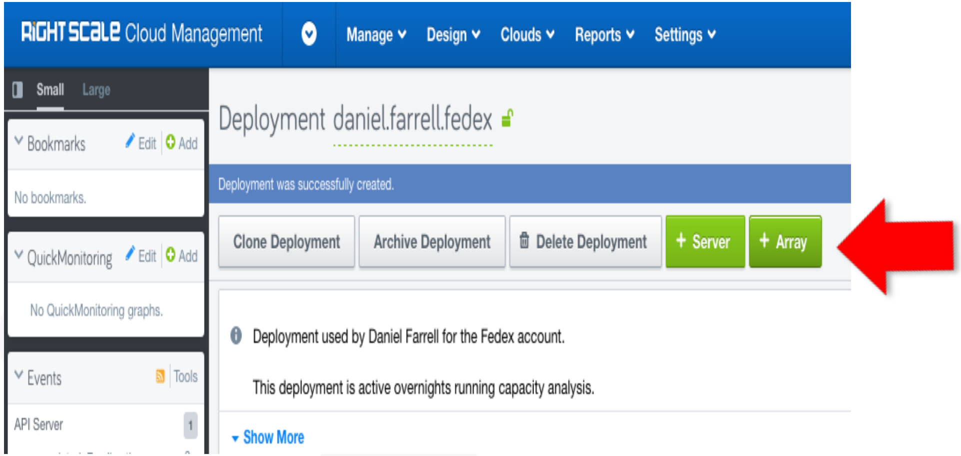
RightScale terminology: Deployment

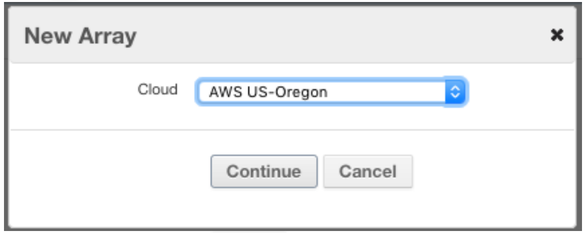
- RightScale is a multi-cloud provider, with which you can deploy nodes (virtual machines, servers) *across* AWS, GCE, Azure, others. We're not completing this type of deployment today.
- As such, however, RightScale uses terminology different and abstract from any cloud provider.

- A RightScale Deployment is an logical construct consisting of:
  - A number of individual servers, each possibly with a different hardware profile.
  - A number of arrays of servers. Each server array member will have the same hardware profile. Each single array will reside in one (cloud provider availability zone).
  - And a mix and/or match of individual and arrays of servers.

- Today we are creating one array of five servers.

Step 3: Create an array of servers deployment





- Click, +Array
- In the dialog bock that is produced, Select “AWS US – Oregon”
- And Click, Continue

- Do I have to use AWS US – Oregon ? No. However, there are combinations of location, hardware profile, storage options, and related that you can choose that ultimately will not work.
  The path we are detailing here works.
  Even in one AWS region we can emulate global network latencies and such using Linux trickery.

The power behind the moment. | DATASTAX

RightScale terminology: Server Template

- A RightScale Server Template is another abstraction.
- If we refer to a given virtual machine image as an OVF, OVA, or VMDK set of files, each cloud provider has specific and differing sets of (rules, requirements, limitation, whatever) before you can load or make use of same.
- RightScale uses Server Templates to try and aid you in this task.

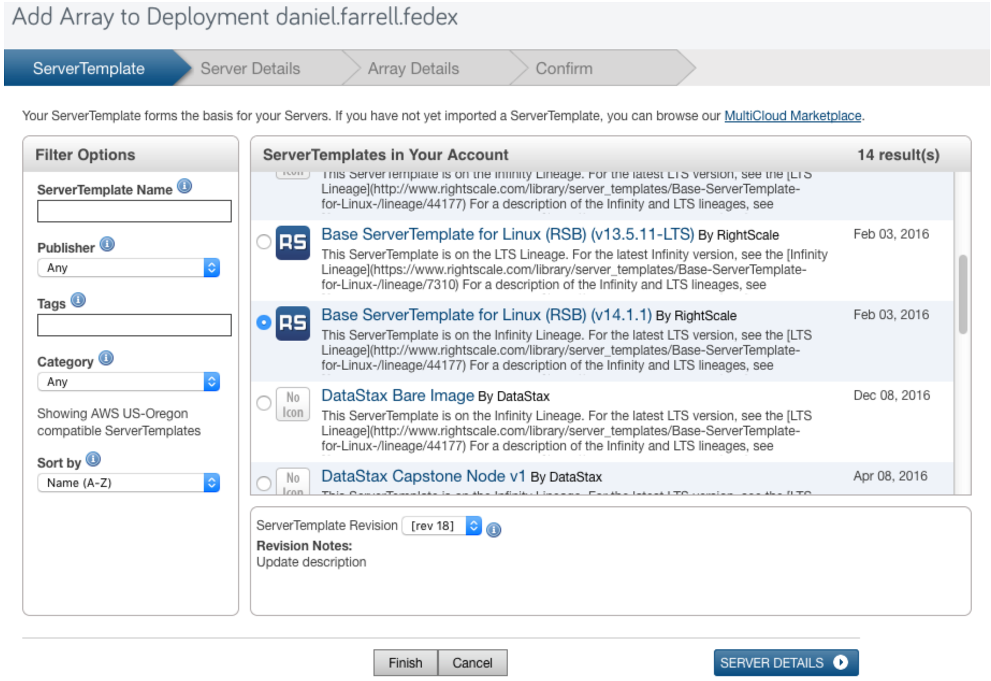
- Net/net, you can use RightScale Server Templates, even customize them, or use brute force. We prefer brute force.

- After we instantiate a given server, if we find we need a given version of Python or whatever on that machine, we'll just script that update or related.

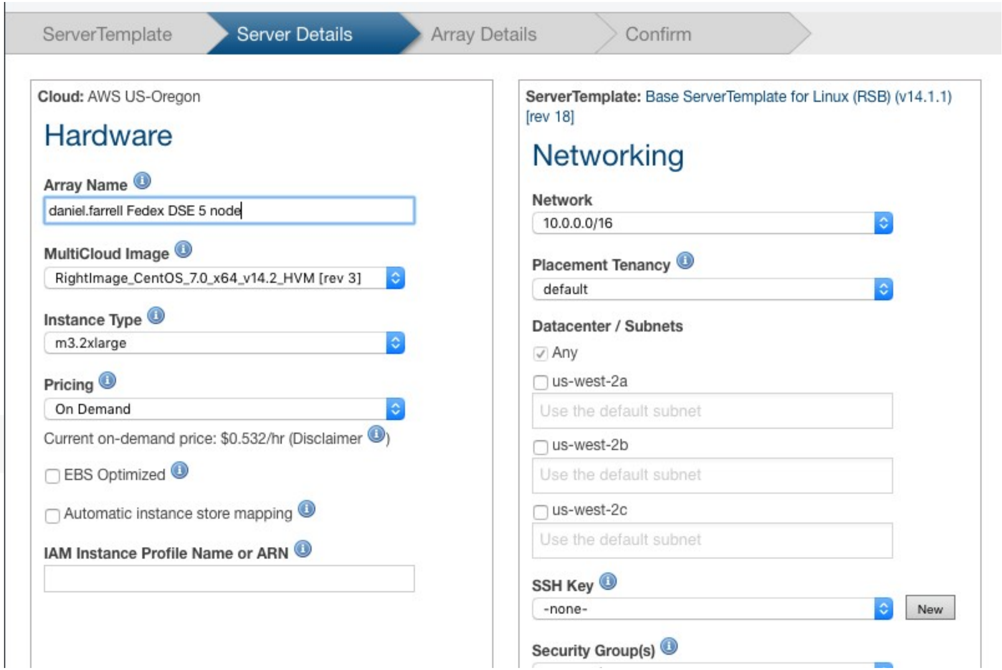
Step 4: Select a server template type

- Select the RightScale Server Template titled, Base Server Template for Linux (RSB) (v14.1.1)
- Then Click, Server Details

- Why use this server template ? Because it has worked for us before, providing us the features we desire on the next and subsequent screens.



Step 5: Server Details



We call to make a number of changes here:

- Edit the (Server) Array Name, to include your personal identifier.

- MultiCloud Image, and Instance Type, see next slide-

Step 5: continued

- For MultiCloud Image we choose,

  RightImage CentOS 7 x64 v14.2 HVM

  Why ? Because we wanted CentOS version 7.
  Why HVM ? Largely because HVM (versus EBS) gives us exactly what we need for an ephemeral DSE system.
  Be advised that HVM or EBS, some AWS machine types are not compatible with both HVM or EBS.

- For (AWS) Instance Type we choose,

  m3.2xlarge

  Why ? The AWS EC2 machine types are listed here,
  https://aws.amazon.com/ec2/instance-types/
  m3.2xlarge gives us 8 (count) vCPUs, 30 GB RAM, and 2 (count) 80GB SSDs, and its HVM compatible.

  Be aware you may need to script formatting, mounting and naming any drives, yadda, after your virtual machine is created.

Step 5: continued, continued

- We've never had success using Automatic instance store mapping, so we'll skip this feature.
- Datacenter / Subsets
  - If you want a DSE system to honestly test RAC placement, DC placement or similar, you will have to make multiple server arrays, each in a different (cloud provider host location, aka availability zone). We are not doing that type of work today, and will leave this entry Checked as, Any.
- SSH Key
  - We need to generate an SSH Key so that we may later login to any of these boxes remotely and safely.
  - Click, (SSH Key) New, give the key a name, and Click, Save. This drop down list box will then automatically populate.

RightScale terminology: Security Group

- A RightScale Security Group determines the firewall rules your virtual machine will be created with.

- Previously, someone at DataStax created the “automaton (default)” Security Group which meets our needs. (In effect, all of the ports are open.)

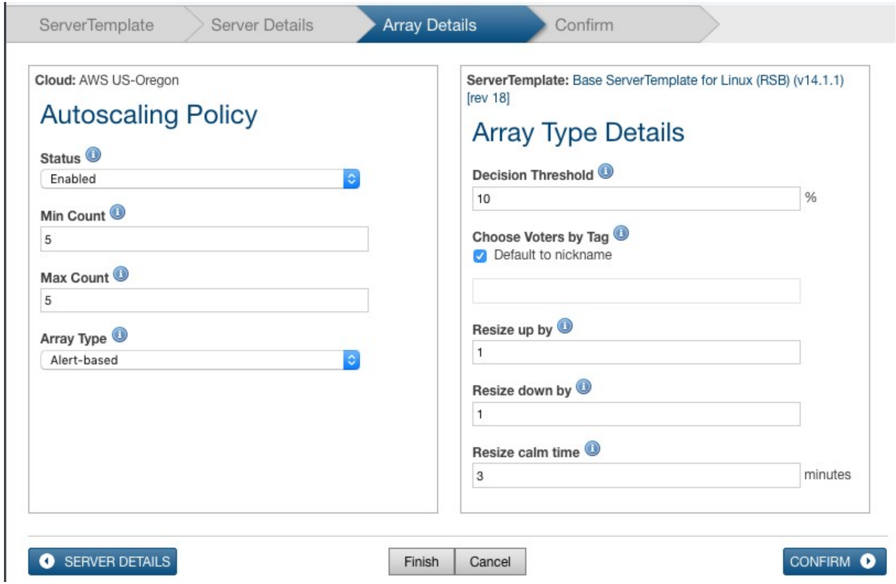
Step 5: final

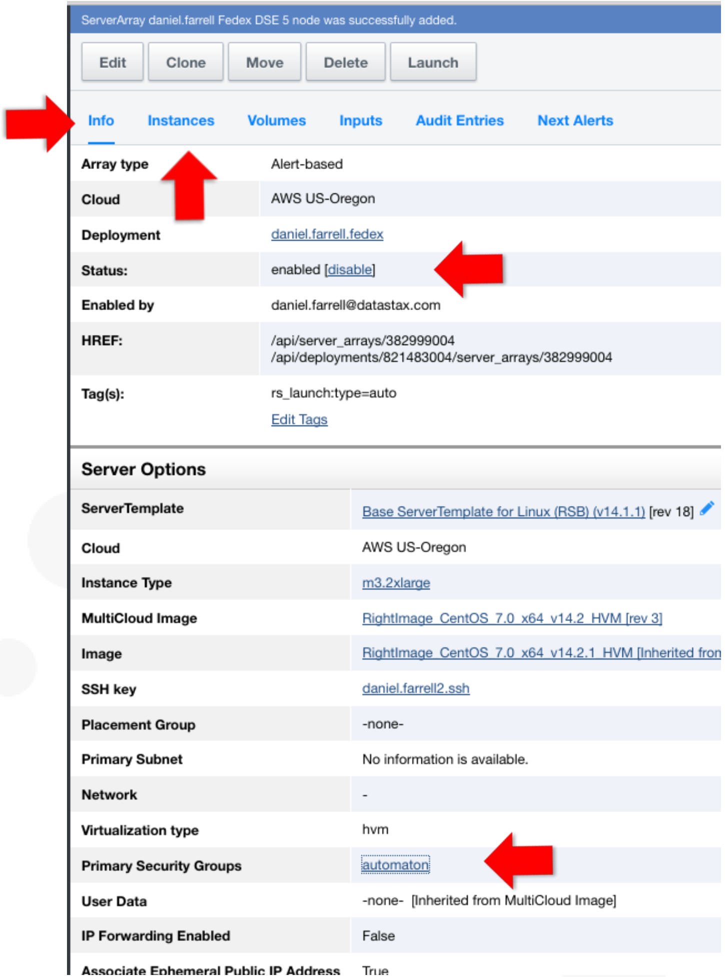
- In the drop down list box for Security Groups, Select, ‘automaton (default)’.
- Leave the Checkbox titled, “Associate Ephemeral Public IP Address” checked.
- And Click, Array Details.

Step 6: Number of servers, and more.

- Change the drop down list box titled, Status, to Enabled.
  If set to Disabled, your virtual machines will not be provisioned.
- Set the Min and Max Count(s) to 5, because we want 5 servers.
- Set Decision Threshold to 10%.
  A rounding error in how this software works; if this number is too high, we have seen where the last server is not provisioned.
- Set Resize calm time to 3 minutes.

- And Click, Confirm -> Finish -> Launch. (Launch will be below the toolbar.)





Result of Step 6, and getting around:

- This screen is produced as the result of our previous step, Step 6. You can return to this screen at any time via the Info button.

- The 'Status enabled|disabled' selection will have to be changed to disabled before we can destroy this server array.
- The 'Primary Security Groups automaton' link shows us our firewall rules, as displayed in the next slide.

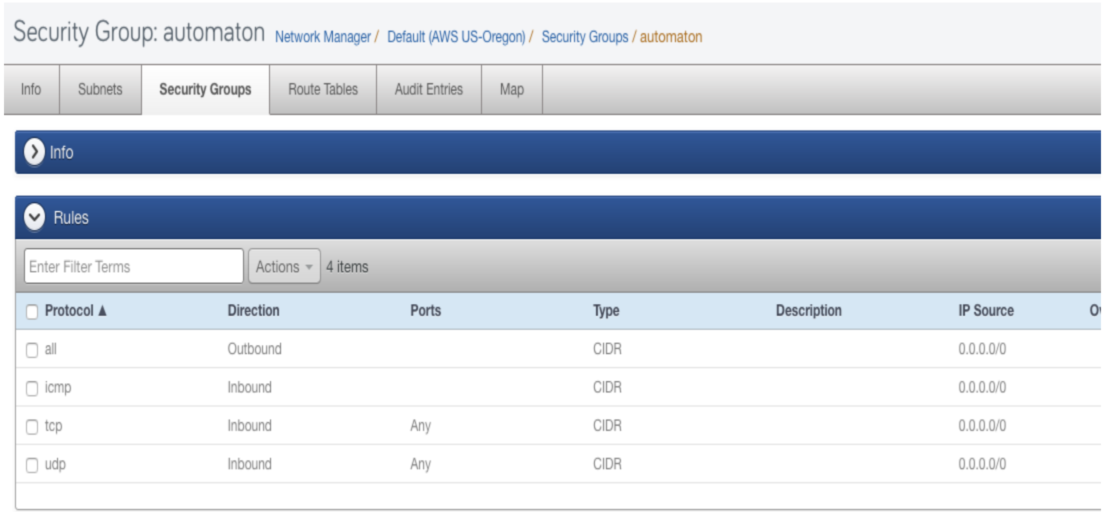
- The Instances button takes us to our server array status screen, as displayed in 2 slides.

Security Group (firewall rules): checking

Viewing our specified firewall rules-

- We get to this screen via the Info button, Primary Security Groups link.

- icmp will give us ping into these boxes.
- Otherwise we are wide open for all inbound and outbound TCP/IP.



Server array status screen:

Nickname	Resource ID	ServerTemplate	Cloud	Instance Type	Started At ▽	CPU	State	Zone	Tags	Runtime	Actions
daniel.farrell Fedex DSE 5 node #5	i-0d6f95254d074cecd	Base ServerTemplate for Linux (RSB) (v14.1.1) [rev 18]	AWS US-Oregon	m3.2xlarge	2017-08-09 12:48:26 PDT		operational	us-west-2b	ec2:Name=daniel.farrell Fedex DSE 5 node #5 provides:rs_agent_type=right_link provides:rs_agent_version=6.3.3 rs_launch:type=auto rs_login:state=restricted rs_monitoring:state=active server:private_ip_0=172.31.38.132 server:public_ip_0=34.208.128.8 server:uuid=04-7NBF077TFK11B	6 minutes	
daniel.farrell Fedex DSE 5 node #4	i-0625ab846efb4fb6f	Base ServerTemplate for Linux (RSB) (v14.1.1) [rev 18]	AWS US-Oregon	m3.2xlarge	2017-08-09 12:48:23 PDT		operational	us-west-2b	ec2:Name=daniel.farrell Fedex DSE 5 node #4 provides:rs_agent_type=right_link provides:rs_agent_version=6.3.3 rs_launch:type=auto	6 minutes	

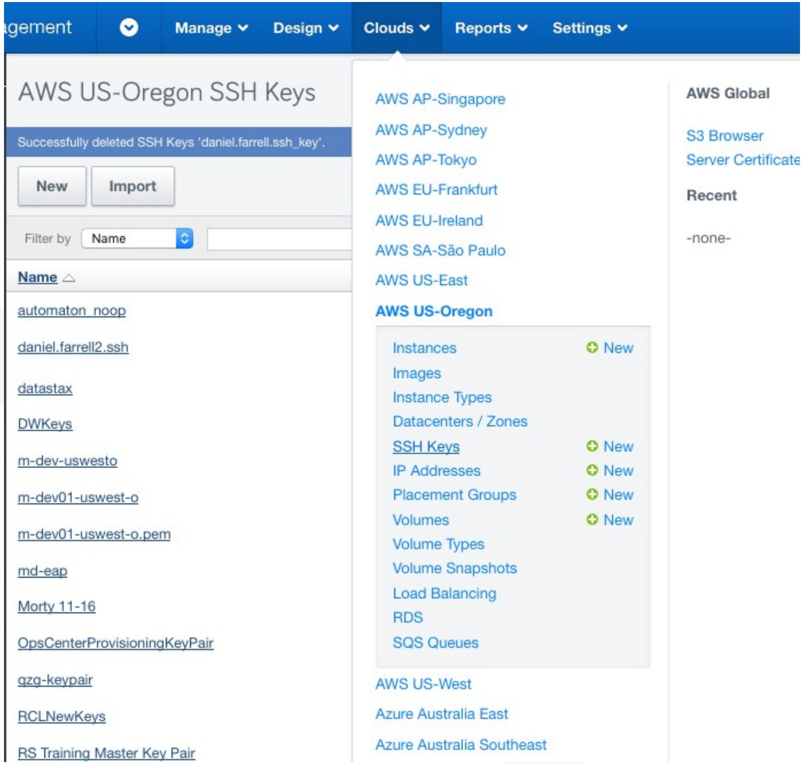
- We arrive at this screen via the Instances button. 5 servers, we have cut the displayed image to afford higher image resolution.
- We see that our boxes are operational. It takes several minutes to move from Launch to Operational.
- Minimally here we need to grab the ‘public IP address’ for each of the 5 servers. This address is in the Tags column for each server.

Status, next steps:

- Thus far we have created a single array of 5 servers. As a single array, these 5 boxes are all of one machine type and reside in one (cloud provider availability zone).

- Next, we need to login into any or all of these boxes to perform our expected work; install and operate DSE or similar.
  - We will need a copy of our SSH Key.
  - We will need the public IP address of each of these boxes.
  - We are going to login using ssh(C) from our laptop into these boxes.

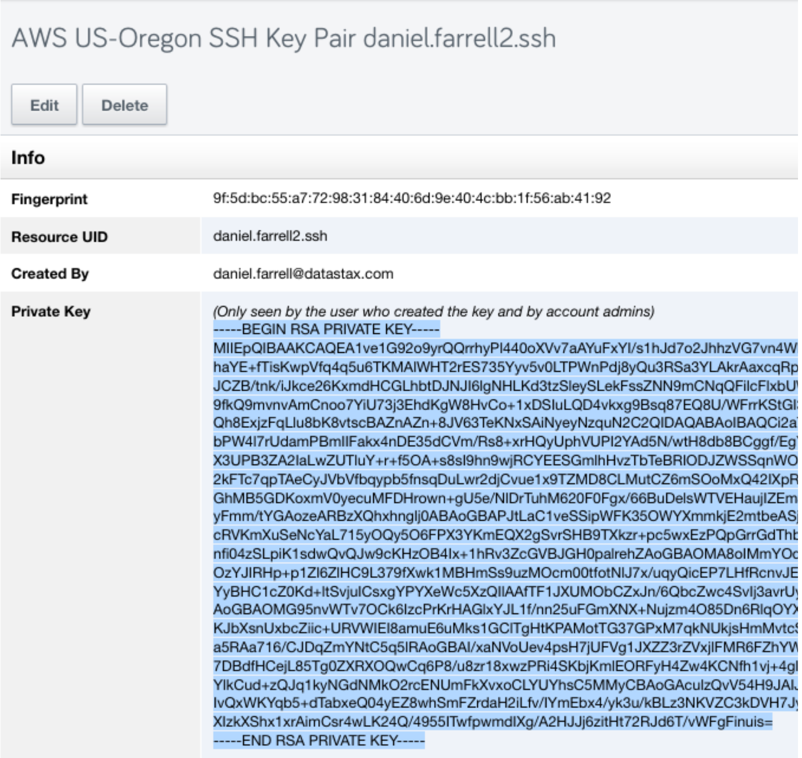
Logging into box 1: Getting the SSH Key



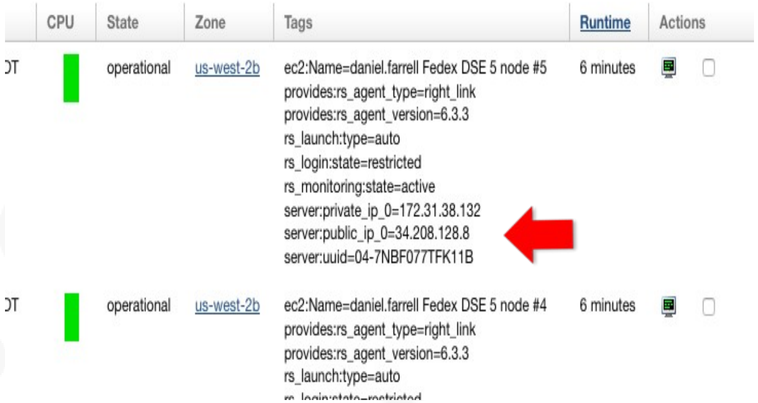
- In Step 5 earlier we generated an SSH Key.
- We can retrieve this value from a, Menu Bar -> Clouds -> (our cloud) -> SSH Keys
- Click, SSH Keys, then Click the named key we created.

Logging into box 1: Copying the key value

- Copy all of the text including the BEGIN RSA and END RSA lines.
- Open any text editor, paste, and save as a named file of your choosing. A .key suffix is optional, but standard. We called our file, zzz.key



Logging into box 1: Get the public IP address

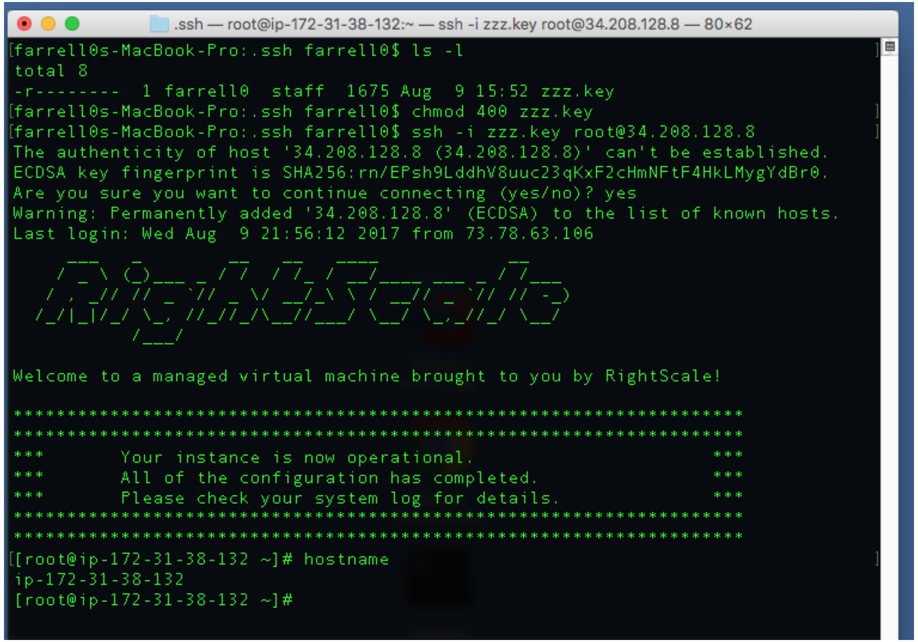


- Recall that our server array public IP addresses are available under, Manage -> Deployments -> (your server array name) -> (your server array name) -> Instances -> Tags -> server:public_ip

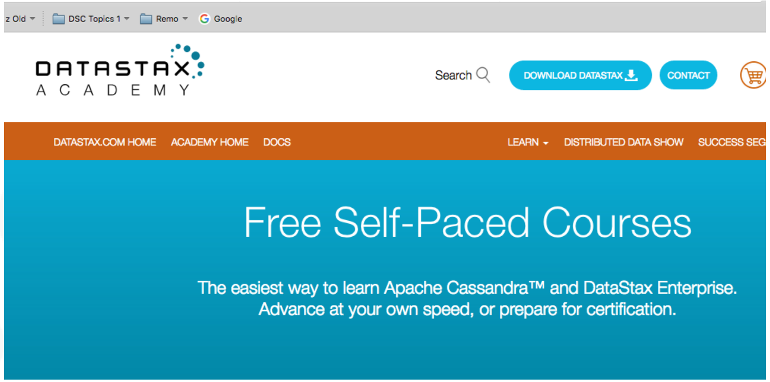
- Capture this IP address.

Logging into box 1: ssh

- From a command window on your laptop, we will need to run ssh, the secure/encrypted shell. On Mac or Linux, this command should be pre-installed.
- The key file you created previously must not be publicly readable. On Mac or Linux we change permissions using a chmod 400 command, as displayed at right.
- Then we perform a,
  ssh –i zzz.key root@34.208.128.8
  Assuming our public IP address is
34.208.128.8



Go forth and prosper-

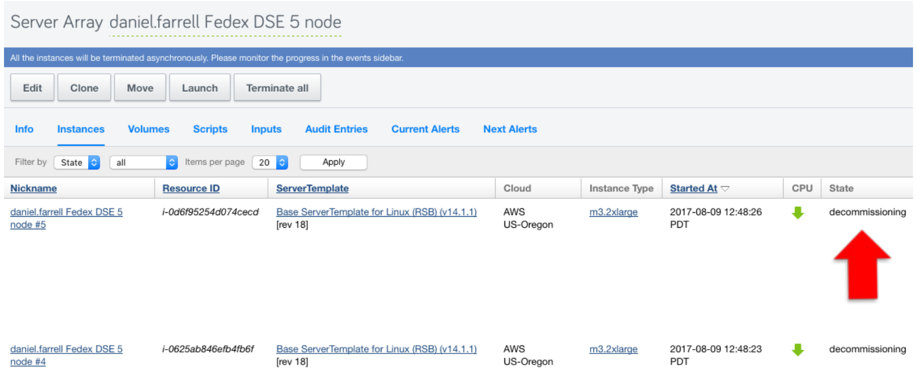


- Now you're ready to do whatever you came to do; install DSE, break DSE, yadda.
- As largely empty server boxes, most folks create a single script that automates all of the settings they prefer:
  - Download, configure and boot DSE
  - Install any client drivers you prefer
  - Other

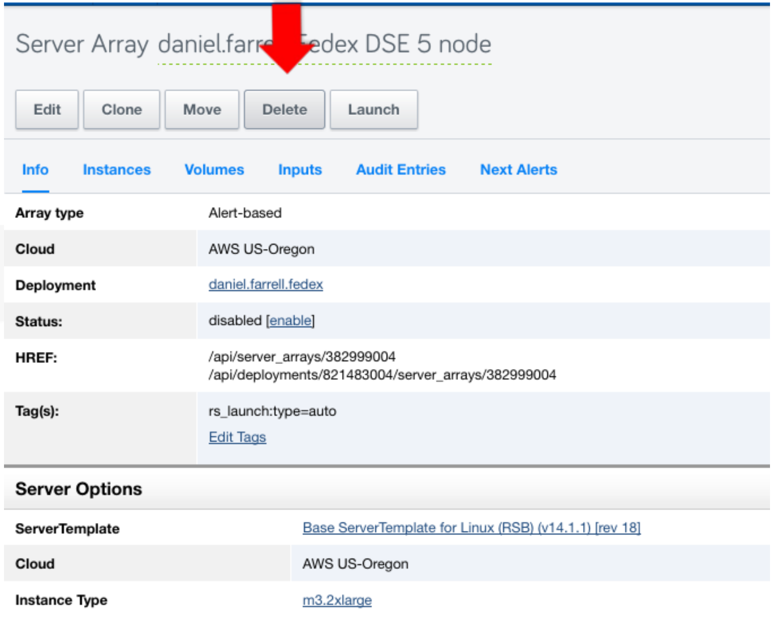
Shutting this all down:

By the nature of the work we perform, it is generally expected that you will un-provision any boxes you created once your work is complete.

- Go to the Info TAB and ensure that Status is disabled, else RightScale will be configured to resurrect your boxes when you try to terminate them.
- Then go to Instances -> Terminate All. There is a safety prompt, then you’re committed.
- Like provisioning, the terminate step may take a few minutes.



Shutting this all down: Confirming



- If you wish to be hyper-safe, we actually go back and delete the now empty (un-provisioned) server array.

Just CTool:

CTool

- Datastax built automation
- Test Engineering “Toolkit”
- Command Line with granular control of environment
- Ability to reset environment for repetitive testing

```
rpmMac:~ rmurphy$ ctool
Usage: ctool [options] command [options] [arg] ...

Tool for performing operations on a cluster. Available operations are:
add_dense_node, automatonize, add_key, activate, bitrock, benchmark_driver
bootstrap, build_packages, create, change_config, change_xml, configure,
cleanup, describe, destroy, docker_machine, dump_cache, dump_key, exec,
get_packages, info, install, install_agents, install_sut, install_only,
java_install, launch, list, list_instances, list_platforms, list_repos, me
metrics, module, perf_monitoring, nose, park, prepare, push, reset, restar
run, run_alias, schedule_shutdown, scp, secure, setup_existing, share_info
snapshot, ssh, ssh_add, start, start_agents, stop, stop_agents, stripe,
sync_cache, tokens, uninstall, unpark, update_repo, upgrade_product and
preflight_check. Run ctool <operation_name> -h to see usage details for an
operation.

Options:
  -h, --help            show this help message and exit
  -r RIGHTSCALE_REGION_NAME, --region-name=RIGHTSCALE_REGION_NAME
                        Region to use. Defaults to default-region-1
  -d, --automaton-dev-mode
                        Turn on dev mode
  -w SSH_WORKER_POOL_SIZE, --ssh-worker-pool-size=SSH_WORKER_POOL_SIZE
                        Sets the number of workers to use for parallel ssh
                        operations on the cluster
  --provider=CLUSTER_PROVIDER
                        The cluster provider you want to use. defaults to
  --log-dir=LOGGING_DIR
                        Use this option to provide an alternative log
                        directory
  --log-file=LOGGING_FILE
                        Use this option to provide an alternative log file
                        name
  -s, --log-stdout      Also log messages to stdout
```

CTOOL (cont)

- Tool to automate the setup of clusters
  - Originally created and used by the test team
  - Open to the wider company
- Riptano repository
  - https://github.com/riptano/ctool
    - what you should be using now
- Pull down to install
  - Make sure to view the readme file
  - For full documentation see:
    - https://datastax.jira.com/wiki/display/QA/CTOOL+DOCUMENTATION
    - https://datastax.jira.com/wiki/display/QA/CTOOL+New+Repo

CTOOL Requirements

- Python 2.7.5 +
- On mac use homebrew to install
  - http://www.howtogeek.com/211541/homebrewforosxeasilyinstallsdesktopappsandterminalutilities/
  - https://coolestguidesontheplanet.com/installinghomebrewonosxelcapitan1011packagemanagerforunixapps/
- DO NOT use mac installed python
- Homebrew install will install pip
- If get errors when first running ctool may need to install additional modules with pip
  - Pip install pythonnovaclient pythonkeystoneclient funcsigs wrapt netifaces==0.10.3 positional monotonic

Create .automaton.conf in Your Home Directory

```
[cluster]provider = rightscale

[rightscale]
# for the private key: login to RightScale, download SSH private key from:
https://us4.rightscale.com/global/users/ssh#ssh,
# save it locally, then set the local path here
user_email = <rightscaleemail>
user_pass = <rightscalepassword>
private_key = /Users/example/bin/keys/example_rightscale_private_key_rsa

# supported cloud providers: ec2, gce
cloud_provider = gce

# account_id as seen in your RightScale url: https://us4.rightscale.com/acct/<account_id>/
account_id = 00000

[logging]
# log file location must be a full path ("~" not allowed)
level = INFO
file = /Users/example/tmp/automaton.log

[credentials]
github_private_key_path = /Users/example/bin/keys/ctool_private_key_rsa
artifactory_user = qaautomaton
artifactory_pass = AKCp2WX2wS8qiMzEmaCureoZ3Tx95B2mTDXfBZLZ6yrYqMdWi6thHX4fuVQGwUDCkoJSde84T

[install]
repo_username = <DSA user name (not email)>
repo_password = <DSA password>
```

Basic CTOOL Use

```
ctool launch p ubuntu MyCluster i m3.2xlarge 2
ctool install b "5.0.0rc2" k 1 s 1 \
   MyCluster enterprise enablegraph
ctool start MyCluster enterprise
```

Getting Info

ctool list

ctool info murphyEAP

```
rmurphy|>>>ctool list
murphyEAP, 2 instances
rmurphy|>>>ctool info murphyEAP
cluster name: murphyEAP
Node 0:
        public hostname: ec2-52-53-190-54.us-west-1.compute.amazonaws.com
        public ip: 52.53.190.54
        private hostname: 172.31.16.180
        private dns: ip-172-31-16-180.us-west-1.compute.internal
        region: EC2 us-west-1
Node 1:
        public hostname: ec2-52-53-173-60.us-west-1.compute.amazonaws.com
        public ip: 52.53.173.60
        private hostname: 172.31.18.96
        private dns: ip-172-31-18-96.us-west-1.compute.internal
        region: EC2 us-west-1
```

Cluster ‘type’

- You can specify what parts of DSE to turn on when defining your cluster

```
e PERCENT_SEARCH
g PERCENT_ANALYTICS
k PERCENT_SPARK
```

  -- PERCENT_* Must be a valid decimal number between 0.0 and 1.0
  -- Where 0 is off and 1.0 is 100%

- To turn on graph simply use

```
enablegraph
```

- To enable DSEFS

```
enabledsefs
```

Adding Complexity

- Datacenter fun:
  ctool launch multiregion 2 "ec2:uswest1:1 ec2:useast1:1"
- Security:
  -- ctool secure kdc –k
  -- https://datastax.jira.com/wiki/display/QA/CTOOL+Secure
- Park/Unpark (Currently park works on EC2 using the "rightscale" cluster provider.)
  -- ctool park superCluster
  -- ctool unpark
  -- https://datastax.jira.com/wiki/pages/viewpage.action?pageId=81592724

Accessing your Servers

You can access your nodes in a couple of ways

- Via ctool directly
  `ctool ssh exampleEAP 0`
- Or using the ssh key to ssh directly as user automaton
  --You will need to put your key into a file with read only permissions for user
  `ssh i myClusterKey automaton@<external ip>`
  -- This is the same method used if accessing any cloud based server though the name may change depending on the template
- To retrieve your ID ssh key use
  `ctool dump_key MyClusterName`

Command Line

You can execute command line actions using ctool to any and all nodes in your cluster

```
ctool run MyCluster all 'sudo mv /home/automaton /mnt1 && \
    sudo ln s /mnt1/automaton /home/’
```

```
ctool run MyCluster all 'sudo sed i
s/"^authenticator: .*$"/"authenticator: com.datastax.bdp.cassandra.auth
.DseAuthenticator"/ /etc/dse/cassandra/cassandra.yaml'
```

Building environments quickly

With CTOOL you can

- SCP
- SSH
- EXEC
- JAVA_INSTALL
- RESET (!!!)
- START/STOP AGENTS
- UPGRADE
- ETC

Getting Help-

- ‘ctool’ by itself will give you a list of commands
- Once you know the commands you can then get help on them individually by using
  -- ctool <command name> help
  -- Or ctool <command name> h
- For example:
  -- ctool install –help
  -- ctool start –h
  -- ctool list help

Final on CTool:

CTOOL is great for doing things in house

- Test something a customer is doing
- Try something new for yourself
- General experimentation

And RightScale is the cloud provider manager behind the scenes

- Used by CTOOL
- Used by Asset Hub
- Can build clusters based on templates

But what if you want to demo for a client how to easy it is to build a cluster on the cloud

- You don't want to use our internal tools
- Because you want to demonstrate in a way they may do it themselves

Final on CTool:

While AWS is very common some customers prefer using other providers

- Maybe do to competition with Amazon
- Maybe because they want to have a hybrid cloud strategy
- Maybe because of previous relations with one provider over the other

We will not do an install on all providers available but do know

- CTOOL and RightScale can provision in AWS or GCE
- We have a Strong Partner with Azure
- DMC currently uses Azure and AWS
- DMC is looking to add either Oracle Cloud or GCE in the near future

No matter which provider you choose the DSE steps are the same and provisioning is based upon the individual console

Just Power Tools, EBDSE:

Power Tools

The Vanguard group has been working on a number of tools for internal consumption

- We have already spoke some about AssetHub
- We next want to introduce EBDSE
  -- Engine Block for DSE
  -- Internal tool used for testing all aspects of DSE not just the Core as per cassandrastress
- Slack channel can be found at
  -- #ebdse
- Documentation can be found at
  -- https://powertools.datastax.com/intro/powertools/
  -- Use academy login to access

Introduction to EBDSE

Cassandra Stress is very useful to become familiar with

- Part of toolset that comes with Open Source Cassandra
- Clients may be familiar with or even already using
- Thus they may have questions about it

But it doesn’t

- Have user friendly data
- Work well for testing Search
- Hard to make useful data for Analytics tests
- Can’t test DSE graph loads

So an internal tool has been developed

- EBDSE

EBDSE

Is an internal ***only*** toolset

- You can’t leave a copy of it at a customer site
  -- Of course you can leave the results of the test, the metrics and such, just not the tool itself
- It is a carrot for sales and services to unfold our product
- We want to protect ourselves from having an internal tool used against us by competitors

Initially designed to enhance the sales cycle

- Be able to do a quick demo
- Rapidly showcase a Proof of Technology (POT)
  -- SE’s may want to note that fact for their project
  -- Many AssetHub assets use EBDSE for testing within
- Demonstrate a user experience that will *scale*

EBDSE

Core Assumptions around design include

- Our customers want to be compelled by facts and evidence, not by ideas alone.
- DataStax and our Customers can't afford to spend weeks or months building a custom testing harness for each opportunity just to establish product fit, relevancy, or scale. Establishing value should be as loweffort as we can make it.
- Analytic methods for establishing workload sizing are moot without data from real tests. Given the choice, real benchmark data is preferred to estimates.

Do simple things fast!

- Simple things should be simple. Complex things should be possible.

So What is EBDSE

A *single* scriptable testing system that

- Creates human readable data
- Can validate a database design
- Can provide load metrics
- Test systems using CQL
- Test systems using DSE graph
- Because of the above two also
  -- Tests out DSE Search
  -- Provides Data to test DSE Analytics
    - o Does not write jobs for you
- Is extensible
  -- Internal tool that is tracked in Jira where you can put in bug and feature requests
- 90% of the time you will not have to do any scripting
  -- Just custom yamls that reflect your data model

How do I get it?

EBDSE is documented at

- https://powertools.datastax.com
  -- Uses your DataStax academy login
  -- Contact Academy team if you can’t get access
  -- Also can get help on the command line
    - ./ebdse help
    - ./ebdse advancedhelp

Has a slack channel for questions and comments

- #ebdse

Download at

- https://powertools.datastax.com/assets/
- curl O https://powertools.datastax.com/assets/ebdse u <dsa account>

Core Machinery

PowerTools Core

- EngineBlock
  - Provides the core runtime
  - Docs http://docs.engineblock.io/
- VirtData
  - Provides the generation of data
  - Docs available at http://docs.virtdata.io/
- Github
  - https://github.com/riptano/ebdse
  - https://github.com/engineblock/engineblock/
  - https://github.com/virtualdataset/virtdatajava

How do I run?

Against default test keyspace (testks) and schema (telemetry)

- Download ebdse
- Execute
  - `./ebdse v run type=cql yaml=telemetry/telemetryschema.yaml host=localhost`
    - This creates the keyspace and schema
  - `./ebdse v run alias=ops type=cql yaml=telemetry/telemetryops.yaml host=localhost cycles=1000`
    - This runs a test with 1000 cycles, feel free to change the numbers
  - Read the results
    - Some results on command line
    - Detailed Output in scenario*.log
- https://powertools.datastax.com/book/intro/quickstart/
- Example yaml files
  - https://github.com/riptano/ebdse/tree/master/ebdse/src/main/resources/activities

Look at the command help for cql

./ebdse help cql

- Note all the options
- Note the example yaml file for a custom table setup